

FLRW Tensor Calculations

Metric Tensor

$$g_{\mu\nu} = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & a^2(t) & 0 & 0 \\ 0 & 0 & a^2(t) & 0 \\ 0 & 0 & 0 & a^2(t) \end{bmatrix}$$

Christoffel Symbols

$$\Gamma_{11}^0 = a(t) \frac{d}{dt} a(t)$$

$$\Gamma_{22}^0 = a(t) \frac{d}{dt} a(t)$$

$$\Gamma_{33}^0 = a(t) \frac{d}{dt} a(t)$$

$$\Gamma_{01}^1 = \frac{\frac{d}{dt} a(t)}{a(t)}$$

$$\Gamma_{02}^2 = \frac{\frac{d}{dt} a(t)}{a(t)}$$

$$\Gamma_{03}^3 = \frac{\frac{d}{dt} a(t)}{a(t)}$$

Riemann Tensor

$$R_{101}^0 = a(t) \frac{d^2}{dt^2} a(t)$$

$$R_{202}^0 = a(t) \frac{d^2}{dt^2} a(t)$$

$$R_{303}^0 = a(t) \frac{d^2}{dt^2} a(t)$$

$$R_{001}^1 = \frac{\frac{d^2}{dt^2} a(t)}{a(t)}$$

$$R_{212}^1 = \left(\frac{d}{dt} a(t) \right)^2$$

$$R_{313}^1 = \left(\frac{d}{dt} a(t) \right)^2$$

$$R_{002}^2 = \frac{\frac{d^2}{dt^2} a(t)}{a(t)}$$

$$R_{112}^2 = - \left(\frac{d}{dt} a(t) \right)^2$$

$$R_{323}^2 = \left(\frac{d}{dt} a(t) \right)^2$$

$$R_{003}^3 = \frac{\frac{d^2}{dt^2} a(t)}{a(t)}$$

$$R_{113}^3 = - \left(\frac{d}{dt} a(t) \right)^2$$

$$R_{223}^3 = - \left(\frac{d}{dt} a(t) \right)^2$$

Ricci Tensor

$$R_{00} = -\frac{3\frac{d^2}{dt^2}a(t)}{a(t)}$$

$$R_{11} = a(t)\frac{d^2}{dt^2}a(t) + 2\left(\frac{d}{dt}a(t)\right)^2$$

$$R_{22} = a(t)\frac{d^2}{dt^2}a(t) + 2\left(\frac{d}{dt}a(t)\right)^2$$

$$R_{33} = a(t)\frac{d^2}{dt^2}a(t) + 2\left(\frac{d}{dt}a(t)\right)^2$$

Ricci Scalar

$$R = \frac{6\left(a(t)\frac{d^2}{dt^2}a(t) + \left(\frac{d}{dt}a(t)\right)^2\right)}{a^2(t)}$$

Einstein Tensor

$$G_{00} = \frac{3.0\left(\frac{d}{dt}a(t)\right)^2}{a^2(t)}$$

$$G_{11} = -2.0a(t)\frac{d^2}{dt^2}a(t) - 1.0\left(\frac{d}{dt}a(t)\right)^2$$

$$G_{22} = -2.0a(t)\frac{d^2}{dt^2}a(t) - 1.0\left(\frac{d}{dt}a(t)\right)^2$$

$$G_{33} = -2.0a(t)\frac{d^2}{dt^2}a(t) - 1.0\left(\frac{d}{dt}a(t)\right)^2$$